Sun Sun

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Research Interests

I am generally interested in optimization, machine learning, and deep learning applications.

Research Experience

Senior Research Officer 08/2019 –
National Research Council

Adjunct Professor 08/2018 –
The Cheriton School of Computer Science, University of Waterloo

Research Associate 01/2017 – 08/2019
Department of Electrical and Computer Engineering, University of Waterloo

Education

Ph.D., Electrical and Computer Engineering, University of Toronto, Canada 09/2011 - 03/2016

- Supervisors: Ben Liang and Min Dong
- Thesis: Management of electrical grids with storage and flexible loads under high-penetration renewables

M.Sc., Electrical and Computer Engineering, University of Alberta, Canada 09/2009 – 08/2011

- Supervisor: Yindi Jing
- Thesis: Channel training and decodings for multiple-input-multiple-output (MIMO) relay networks

B.En., Electrical Engineering and Automation, Tongji University, China 09/2001 – 07/2005

Publications

1. *GeoCTGAN generates privacy-aware geosynthetic event data*
   Abu Yousuf Md Abdullah, Jane Law, **Sun Sun**, Bing Hu, Anita Layton, and Helen Chen
   Scientific Reports, 2026

2. *ASH: Agents that Self-Hone via Embodied Learning*
   Benjamin Schneider, Xavier Schneider, Victor Zhong, **Sun Sun**
   Neural Information Processing Systems (NeurIPS), 2026

3. *DELBERT-2: Pretrained Fingerprint Language Models for DEL Protein Binder Prediction*
   Bing Xu Hu, **Sun Sun**, Shaik salman basha, Anita Layton, and Helen Hong Chen
   ICML workshops on GenBio and FM4LS, 2026

4. *OBELiX: A Curated Dataset of Crystal Structures and Experimentally Measured Ionic Conductivities for Lithium Solid-State Electrolytes*
   Félix Therrien, Jamal Abou Haibeh, Divya Sharma, Rhiannon Hendley, Leah Wairimu Mungai, **Sun Sun**, Alain Tchagang, Jiang Su, Samuel Huberman, Yoshua Bengio, Hongyu Guo, Alex Hernández-García, and Homin Shin
   Digital Discovery, 2026

5. *Context Forcing: Consistent Autoregressive Video Generation with Long Context*
   Shuo Chen, Cong Wei, **Sun Sun**, Tiancheng SHEN, Ping Nie, Kai Zou, Ge Zhang, Ming-Hsuan Yang, and Wenhu Chen
   International Conference on Machine Learning (ICML), 2026

6. *NeuralOS: Towards Simulating Operating Systems via Neural Generative Models*
   Luke Rivard, **Sun Sun**, Hongyu Guo, Wenhu Chen, and Yuntian Deng
   International Conference on Learning Representations (ICLR), 2026

7. *SED2AM: Solving Multi-Trip Time-Dependent Vehicle Routing Problem using Deep Reinforcement Learning*
   Arash Mozhdehi, Yunli Wang, **Sun Sun**, and Xin Wang
   Transactions on Knowledge Discovery from Data, 2025

8. *Domain Knowledge Infused Conditional Generative Models for Accelerating Drug Discovery*
   Bing Xu Hu, Jong-Hoon Park, Anita Layton, Young-Rae Cho, **Sun Sun**, and Helen Hong Chen
   NeurIPS Workshop on AI Virtual Cells and Instruments: A New Era in Drug Discovery and Development (AI4D3), 2025

9. *DisProtEdit: Controllable Protein Editing via Disentangled Text Representations*
   Max Ku, **Sun Sun**, Hongyu Guo, and Wenhu Chen
   ICML Workshop on Generative AI and Biology (GenBio), 2025

10. *DisProtEdit: Exploring Disentangled Representations for Multi-Attribute Protein Editing*
    Max Ku, **Sun Sun**, Hongyu Guo, and Wenhu Chen
    ICML Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences (FM4LS), 2025

11. *Dray-Q: Demand-Dependent Trailer Repositioning using Deep Reinforcement Learning*
    Hadi Aghazadeh, Yunli Wang, **Sun Sun**, and Xin Wang
    Transportation Research Part C: Emerging Technologies, 2024

12. *Clustering-Based Enhanced Ant Colony Optimization for MultiTrip Vehicle Routing Problem with Heterogeneous Fleet and Time Windows: An Industrial Case Study*
    Beom Sae Kim, Arash Mozhdehi, Yunli Wang, **Sun Sun**, and Xin Wang
    International Workshop on Computational Transportation Science (IWCTS), 2024

13. *EFECTIW-ROTER: Deep Reinforcement Learning Approach for Solving Heterogeneous Fleet and Demand Vehicle Routing Problem With Time-Window Constraints*
    Arash Mozhdehi, Mahdi Mohammadizadeh, Yunli Wang, **Sun Sun**, and Xin Wang
    ACM SIGSPATIAL, 2024

14. *FleetWiz: An Intelligent Platform for Heterogeneous Spatio-Temporal Truckload Fleet Dispatching*
    Saeid Kalantari, Yunli Wang, **Sun Sun**, and Xin Wang
    ACM SIGSPATIAL, 2024 (Demo paper)

15. *DP-LFlow: Differentially Private Latent Flow for Scalable Sensitive Image Generation*
    Dihong Jiang and **Sun Sun**
    Transactions on Machine Learning Research, 2023

16. *Trailer Allocation and Truck Routing Using Bipartite Graph Assignment and Deep Reinforcement Learning*
    Saeid Kalantari, Reza Safarzadeh Ramhormozi, Yunli Wang, **Sun Sun**, and Xin Wang
    Transactions in GIS. 2023.

17. *f-MICL: Understanding and Generalizing InfoNCE-based Contrastive Learning*
    Yiwei Lu, Guojun Zhang, **Sun Sun**, Hongyu Guo, and Yaoliang Yu
    Transactions on Machine Learning Research, 2023

18. *Functional Rényi Differential Privacy for Generative Modeling*
    Dihong Jiang, **Sun Sun**, and Yaoliang Yu
    Neural Information Processing Systems (NeurIPS), 2023

19. *An Interactive Map-based System for Visually Exploring Goods Movement based on GPS Traces*
    Reza Safarzadeh, Yunli Wang, **Sun Sun**, and Xin Wang
    International Symposium on Spatial and Temporal Data (SSTD), 2023. **(Best Demo Paper Award)**

20. *DP-LFlow: Differentially Private Latent Flow for Scalable Sensitive Image Generation*
Dihong Jiang, **Sun Sun**, and Yaoliang Yu
International Conference on Machine Learning (ICML) Workshop on Challenges of Deploying Generative AI, 2023.

21. *Functional Renyi Differential Privacy for Generative Modeling*
Dihong Jiang, **Sun Sun**, and Yaoliang Yu
International Conference on Machine Learning (ICML) Workshop on Challenges of Deploying Generative AI, 2023.

22. *CM-GAN: Stabilizing GAN Training with Consistency Models*
Haoye Lu, Yiwei Lu, Dihong Jiang, Spencer Ryan Szabados, **Sun Sun**, and Yaoliang Yu
International Conference on Machine Learning (ICML) Workshop on Structured Probabilistic Inference & Generative Modeling, 2023.

23. *Solving a Multi-Trip VRP with Real Heterogeneous Fleet and Time Windows based on Ant Colony Optimization: An Industrial Case Study*
Jihee Han, Arash Mozhdehi, Yunli Wang, **Sun Sun**, and Xin Wang
International Workshop on Computational Transportation Science (IWCTS 2022).

24. *Multi-Task Graph Neural Network for Truck Speed Prediction Under Extreme Weather Conditions*
Reza Safarzadeh Ramhormozi, Arash Mozhdehi, Saeid Kalantari, Yunli Wang, **Sun Sun**, and Xin Wang
International Conference on Advances in Geographic Information Systems 2022 (ACM SIGSPATIAL 2022).

25. *Revisiting flow generative models for Out-of-distribution detection*
Dihong Jiang, **Sun Sun**, and Yaoliang Yu
International Conference on Learning Representations (ICLR), 2022.

26. *f-Mutual Information Contrastive Learning*
Guojun Zhang, Yiwei Yu, **Sun Sun**, Hongyu Guo, and Yaoliang Yu
Advances in Neural Information Processing Systems (NeurIPS) workshop on SSL. 2021.

27. *Symmetric Wasserstein Autoencoders*
Sun Sun and Hongyu Guo
Uncertainty in Artificial Intelligence (UAI), July 2021.

28. *Hierarchical Reinforcement Learning for Vehicle Routing Problems with Time Windows*
Yunli Wang, **Sun Sun**, and Wei Li
Canadian AI, May 2021.

29. *Text Style Transfer: Leveraging a Style Classifier on Entangled Latent Representations*
Xiaoyan Li, **Sun Sun**, and Yunli Wang
Workshop on Representation Learning for NLP (REPL4NLP), August 2021.

30. *Multivariate Triangular Quantile Maps for Novelty Detection*
Jingjing Wang, **Sun Sun**, and Yaoliang Yu
Conference on Neural Information Processing Systems (NeurIPS), December 2019.

31. *Using Synthetic Traces for Robust Energy System Sizing*
Sun Sun, Fiodar Kazhamiaka, Srinivasan Keshav, and Catherine Rosenberg
Proc. ACM eEnergy 2019, June 2019.

32. *Least Squares Estimation of Weakly Convex Functions*
Sun Sun and Yaoliang Yu
International Conference on Artificial Intelligence and Statistics (AISTATS), April 2019.

33. *Optimal matching of stochastic solar generators to stochastic loads*
Sun Sun, Srinivasan Keshav, Catherine Rosenberg, and Matthew Peloso
Proc. ACM eEnergy 2018, June 2018.

34. *Phase balancing using energy storage in power grids under uncertainty*
Sun Sun, Min Dong, Ben Liang, and Josh Taylor
IEEE Transactions on Power Systems, vol. 31, no. 5, pp. 3891-3903, Sept. 2016.

35. *Distributed real-time algorithm for renewable-integrated power grids with storage and flexible loads*
Sun Sun, Min Dong, and Ben Liang
IEEE Transactions on Smart Grid, vol. 7, no. 5, pp. 2337-2349, Sept. 2016.

36. *Cost-minimizing distributed algorithm for managing renewable-integrated power grids (invited paper)*
Sun Sun, Min Dong, and Ben Liang
IEEE Global Conference on Signal and Information Processing (GlobalSIP), Dec. 2015.

37. *Distributed real-time phase balancing for power grids with energy storage*
Sun Sun, Josh Taylor, Min Dong, and Ben Liang
American Control Conference (ACC), Jul. 2015.

38. *Real-time power balancing in electric grids with distributed storage*
Sun Sun, Min Dong, and Ben Liang
IEEE Journal of Selected Topics in Signal Processing (special issue on signal processing in smart electric power grid), vol. 8, no. 6, pp. 1167-1181, Dec. 2014.

39. *Real-time welfare-maximizing regulation allocation in dynamic aggregator-EVs system*
Sun Sun, Min Dong, and Ben Liang
IEEE Transactions on Smart Grid, vol. 5, no. 3, pp. 1397-1409, May 2014.

40. *Joint supply, demand, and energy storage management towards microgrid cost minimization*
Sun Sun, Min Dong, and Ben Liang
IEEE Conference on Smart Grid Communications (SmartGridComm), Nov. 2014.

41. *On stochastic feedback control for multi-antenna beamforming: formulation and low-complexity algorithms*
Sun Sun, Min Dong, and Ben Liang
IEEE Transactions on Wireless Communications, vol. 13, no. 9, pp. 4731-4745, Sept. 2014.

42. *Distributed regulation allocation with aggregator coordinated electric vehicles*
Sun Sun, Min Dong, and Ben Liang
IEEE Conference on Smart Grid Communications (SmartGridComm), Oct. 2013.

43. *Real-time welfare-maximizing regulation allocation in aggregator-EVs systems*
Sun Sun, Min Dong, and Ben Liang
IEEE Conference on Computer Communications (INFOCOM) Workshop on Communications and Control for Smart Energy Systems (CCSES), Apr. 2013.

44. *Training and decodings for cooperative network with multiple relays and receive antennas*
Sun Sun and Yindi Jing
IEEE Transactions on Communications, vol. 60, no. 6, pp. 1534-1544, Jun. 2012.

45. *Channel training design in amplify-and-forward MIMO relay networks*
Sun Sun and Yindi Jing
IEEE Transactions on Wireless Communications, vol. 10, no. 10, pp. 3380-3391, Oct. 2011.

46. *Channel training and coherent decodings in amplify-and-forward relay network*
Sun Sun and Yindi Jing
IEEE Global Communications Conference (Globlecom), Dec. 2011.

47. *Channel training and estimation in distributed space-time coded relay networks with multiple transmit/receive antennas*
Sun Sun and Yindi Jing
IEEE Wireless Communications and Networking Conference (WCNC), Apr. 2010.

Co-supervised students

- 04/2025 - present Co-supervise Ph.D. student Bing Hu and Master students YiXin Li and Ping Yen from University of Waterloo
- 04/2025 - 09/2026 Co-supervise Master student Benjamin Schneider from University of Waterloo
- 04/2024 – 04/2026 Co-supervise Ph.D. student Max Ku and Master student Luke Rivard from University of Waterloo
- 02/2023 – 08/2025 Co-supervise Postdoc Salatan Duangdangchote from NRC
- 05/2023 – 12/2023 Co-supervise Master student Karanjot Singh Saggu from University of Ottawa
- 09/2022 – 01/2025 Co-supervise Master student Shawn Kim from University of Calgary
- 09/2021 – 01/2026 Co-supervise 3 Ph.D. students Arash Mozhdehi, Reza Safarzadeh Ramhormozi, and Saeid Kalantari from University of Calgary
- 09/2021 – 01/2023 Co-supervise Postdoc Jihee Han from University of Calgary
- 09/2020 – 09/2023 Co-supervise 3 Ph.D. students Haoye Lu, Yiwei Lu, and Dihong Jiang from University of Waterloo
- 09/2019 – 05/2020 Co-supervise Master student Jingjing Wang from University of Waterloo

Research Funding

- Sun Sun (PI), Helen Chen and Anita Layton from University of Waterloo, projects on "Canada Synthetic Health Data (CAN-SYNH) Network for Health AI Innovation" under NRC-UW collaboration program (2025/04–2027/09). Funding: $200K(CAD).
- Sun Sun (PI) and Wenhu Chen from University of Waterloo, projects on "Towards acceleration of diffusion models in multimodal generation" under New Beginnings program of NRC (2025/04–2026/05). Funding: $50K(CAD).
- Sun Sun (PI) and Wenhu Chen from University of Waterloo, projects on "Accelerating scientific discovery with large language models" under AI4D challenge program of NRC (2024/04–2026/04). Funding: $130K(CAD).
- Sun Sun (PI) and Yuntian Deng from University of Waterloo, projects on "Towards Simulating Operating Systems via Neural Generative Models" under AI4D challenge program of NRC (2024/04–2026/04)
- Sun Sun (co-PI) and Mila, projects on Machine Learning for Efficient and Effective Molecule Search under AI4D challenge program of NRC (2023/09–2026/09). Funding: $1M(CAD)
- Sun Sun (co-PI) and Xin Wang from University of Calgary, projects on AI for logistics of NRC (2021/04–2026/01). Funding: $1.394M(CAD)
- Sun Sun (PI) and Yaoliang Yu from University of Waterloo, projects on deep generative models (2020/09–2024/09) under NRC-UW collaboration program. Funding: $180K(CAD)

Service (Reviewer)

- Round 7 of the New Beginnings Initiative at NRC, 2025
- International Conference on Learning Representations (ICLR)
- Neural Information Processing Systems (NeurIPS)
- International Conference on Machine Learning (ICML)

- Association for the Advancement of Artificial Intelligence (AAAI)
- IEEE Transactions on Smart Grid
- IEEE Transactions on Wireless Communications
- IEEE International Conference on Computer Communications (INFOCOM)

Teaching

- Spring 2018, Instructor for *Introduction to Machine Learning*, University of Waterloo
- Fall 2014, Teaching assistant for *Communication Systems*, University of Toronto
- Fall 2013, Teaching assistant for *Random Processes*, University of Toronto
- Fall 2012, Guest lecturer for *Random Processes*, University of Toronto
- Fall 2010, TA for *Discrete Time Signals and Systems*, University of Alberta